



Bioluminescent protease indicator for drug screen and point-of-care diagnostics

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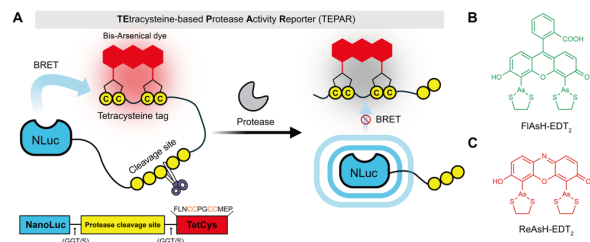
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Abstract

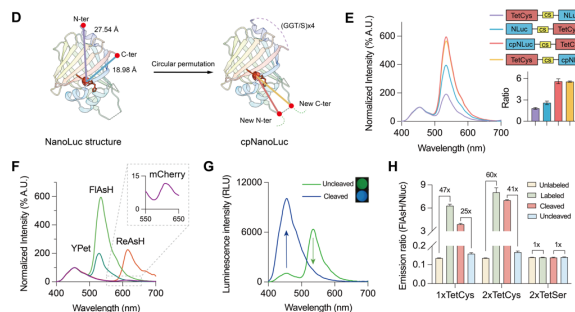
Proteases play a crucial role in essential physiological processes and are recognized as therapeutic targets and diagnostic biomarkers for different pathologies. Here, we developed a new class of Tetracysteine-based Protease Activity Reporters (TePARs) by sandwiching peptides containing protease cleavage sites in between the NanoLuc protein and tetracysteine motif. TePARs used bioluminescence resonance energy transfer (BRET) to shift emitted light from green to red or blue upon proteolysis, allowing detection with either a plate reader or a simple smartphone camera. With further optimization, TePAR 2.0 utilized tandem tetracysteine peptides to achieve unprecedented BRET efficiency (~90%), large dynamic range (>40-fold), and a picomolar limit of detection (LoD). Moreover, TePARs feature a modular design that enables the generation of a wide range of protease sensors by simply swapping peptide substrates for different proteases while maintaining exceptional performance. We used TePAR to quantify active proteases at physiologically relevant concentrations in challenging biological samples, yielding results consistent with those from standard testing methods. Notably, TePARs enable the quantification of MMP-9 plasma levels of colorectal cancer patients at the point of care. TePAR assay distinguished patients with colorectal cancer, including those with early-stage adenomatous polyps, from healthy controls with accuracies of over 90%, thus expanding their utility for point-of-care diagnostics. Furthermore, we also demonstrate that TePAR enables camera-based drug screening and characterization without the need for specialized instruments.

Method

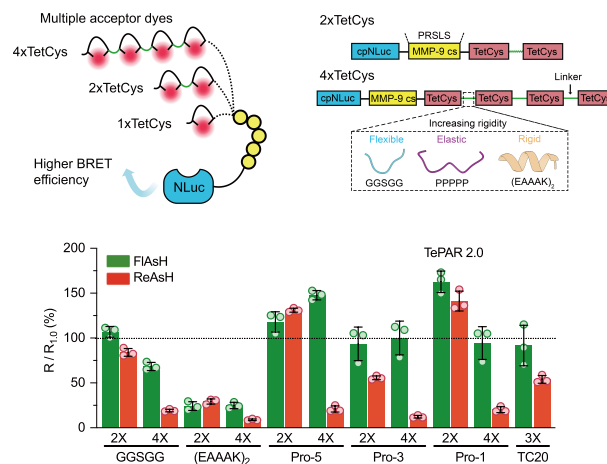


Result

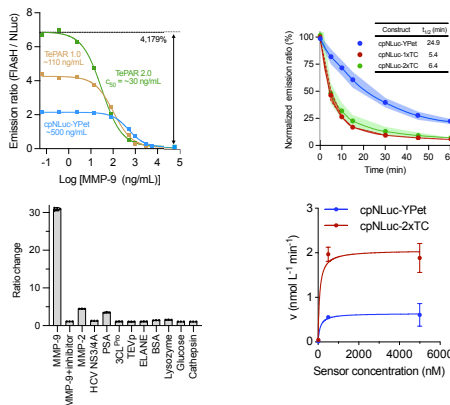
1. Development of an MMP-9 sensor



2. Optimization of MMP-9 sensor



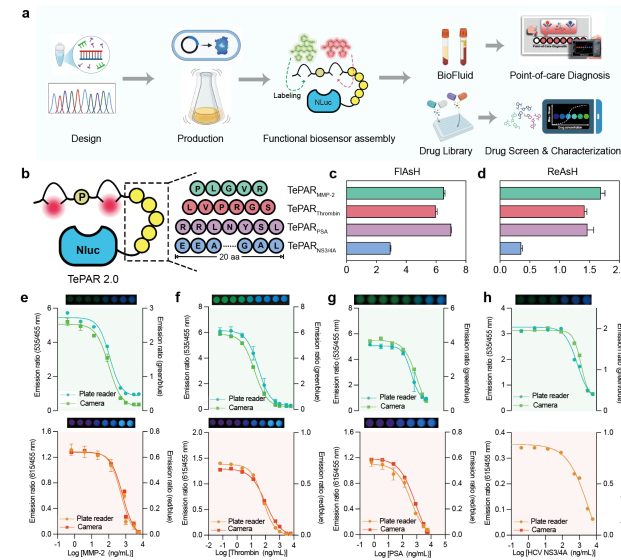
3. Performance of MMP-9 sensor



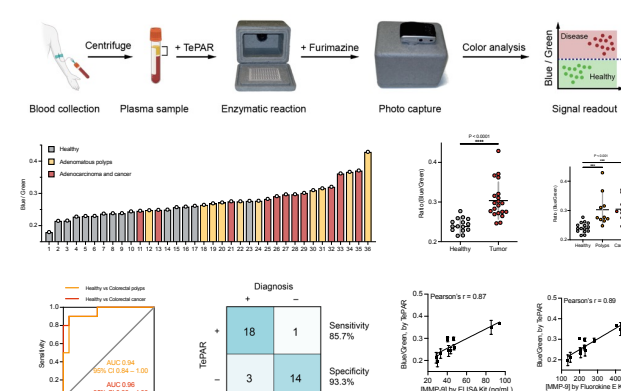
Kinetic constants for MMP-9 sensor

Kinetic constants	NLuc-2xTC	NLuc-YPet
K_m (nM)	76.8	104.4
k_{cat} (min ⁻¹)	3.37	1.05
k_{cat}/K_m (M ⁻¹ s ⁻¹)	7.31×10^5	1.68×10^5
V_m (nM min ⁻¹)	2.06	0.64

4. Modularity of TePAR



5. Clinical sample



This work was supported by